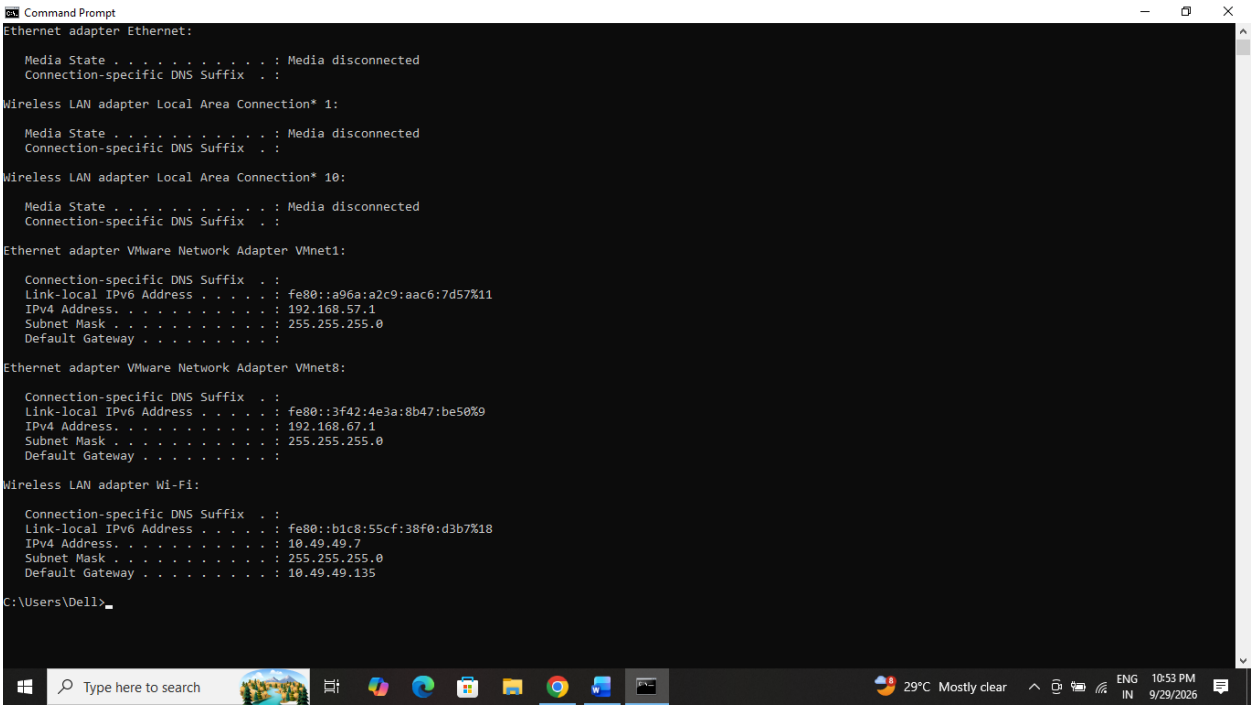
IP addressing

Tasks

1. Find the IP address of your computer or mobile device and write down whether it is IPv4 or IPv6. Also mention if it is a public or private IP.

Ans.:-



If your ipconfig shows an address beginning with 192.168.57.1 it is a private IP.

**2. Given the following list of IP addresses, identify which ones are valid private IPs and which are public: 192.168.1.10, 10.0.0.5, 172.16.5.20, 8.8.8.8, 123.45.67.89.

Hint: Check the standard private IP address ranges for IPv4.**

Ans.:-

IP Address	Type	Reason
192.168.1.10	Private	Falls within 192.168.0.0/16
10.0.0.5	Private	Falls within 10.0.0.0/8
172.16.5.20	Private	Falls within 172.16.0.0/12
8.8.8.8	Public	Not in any private range
123.45.67.89	Public	Not in any private range

**3. Write a short script in Python or JavaScript that takes an IP address as input and tells the user whether it is a valid IPv4 address or not.

Constraint: Do not use any third-party libraries for validation.**

Ans.:-

```
ip = input("Enter an IPv4 address: ")
parts = ip.split(".")
if len(parts) == 4:
    valid = True
    for part in parts:
        if not part.isdigit() or not 0 <= int(part) <= 255:
            valid = False
            break
    if valid:
        print("Valid IPv4 address")
    else:
        print("Invalid IPv4 address")
else:
    print("Invalid IPv4 address")
```

4. Explain with an example how Instagram or WhatsApp might use IP addresses to detect suspicious login activity from different locations.

Ans.:-

Suspicious Login Detection Using IP Addresses

Instagram or WhatsApp can use the IP address associated with a login to identify the approximate network/location from which an account is being accessed. An IP address generally does not provide an exact physical location by itself.

Example:

1. You normally log in from Ahmedabad, India, using an IP address associated with that area.
2. A few minutes later, the same account receives a login from an IP address associated with London, UK.
3. The service may identify this as unusual because the two login locations are far apart in a short period.
4. It may respond with a security alert, request additional verification, or temporarily restrict the login.
5. The system can consider other signals too, such as the device, login time, and previous account activity.